NMR

Nuclear Magnetic Resonance in Liquids

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1 Theoretical Background

Nuclear magnetic resonance (NMR) is a spectroscopic technique used to study atomic nuclei with non-zero spin in a magnetic field. In NMR, nuclei in a sample are exposed to a strong magnetic field and irradiated with radiofrequency radiation, causing transitions between spin states. The resulting spectra provide detailed information about molecular structure, dynamics, and environment, making it a powerful tool in chemistry, biology, and materials science.

1.1 Basics of NMR

1.1.1 Nuclear resonance effect

When a substance containing nuclei with non-zero spin is placed in an external magnetic field, the nuclear spins can align either parallel or antiparallel to the field direction, corresponding to spin quantum numbers$m_I = \pm\frac{1}{2}$ for spin-$\frac{1}{2}$ nuclei. These orientations represent distinct energy states, lifting the degeneracy of the nuclear spin levels. More generally, a nucleus with spin I splits into 2I+1 distinct energy states. This splitting of energy levels due to the interaction between the magnetic field and the magnetic moment of nuclei is known as the *Zeeman effect*.

The energy separation between neighboring energy states is given by:

$$\Delta E_{m_I} = \mu_z B_0 \tag{1}$$

where B_0 is the applied magnetic field, and $\Delta\mu_z = \gamma m_I \hbar$ is the component of the magnetic moment $\boldsymbol{\mu}$ of the nucleus along the field direction. Here γ is the gyromagnetic ratio of the system.

Larmor precession is the precession of the magnetic moment of an object about an external magnetic field. A nucleus with angular momentum given by

$$\mathbf{J} = \frac{1}{\gamma}\mu$$

experiences a torque

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$$

This makes the angular momentum $\mathbf{J}$ process around the magnetic field direction with the angular frequency [2]

$$\omega = \gamma B_0 \tag{2}$$

which is known as the *Larmor frequency*.

The gyromagnetic ratio γ, with units of $rad \cdot s^{-1} T^{-1}$, indicates how rapidly the spin axis of a nucleus precesses per tesla of magnetic field strength. It is determined by intrinsic properties of the nucleus such as its mass, charge, and spin, and is also influenced by the strength and orientation of the external magnetic field. The sign of γ determines the sense of the precession: clockwise or counterclockwise (see Fig. 1).

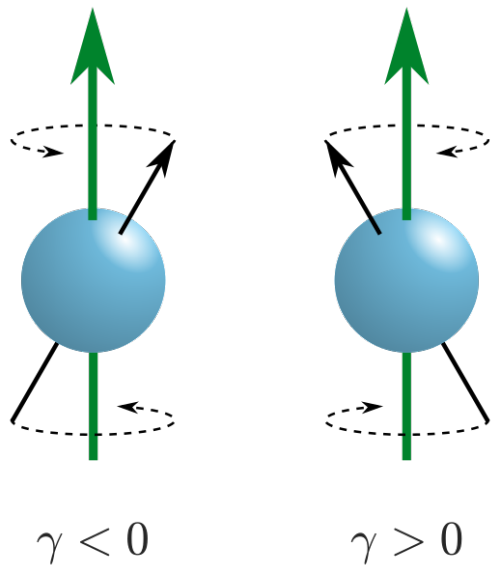


Figure 1: Change of precession direction based on γ [2]

1.1.2 Equilibrium magnetization

For NMR we usually consider the contribution of all non-zero spin nuclei in the sample, obtaining the net (macroscopic) magnetization

$$\mathbf{M} = \frac{1}{V} \sum_i \boldsymbol{\mu_i}$$

where V is the volume of the sample that we are considering and $\mu_\mathbf{i}$ are the individual magnetic moments of the nuclei.

Each energy level due to Zeeman splitting has a different population (N_+, N_-), which is related to the energy difference by the Boltzmann distribution:

$$\frac{N_+}{N_-} = e^{\Delta E/k_B T} = e^{\gamma \hbar B_0/k_B T} \tag{3}$$

where T is the temperature and k_B is the Boltzmann constant. This is generally a value close to but larger than one, which results in a small net magnetization along the external magnetic field direction (as anti-aligned spins are the higher energy state), precessing around the external field direction.

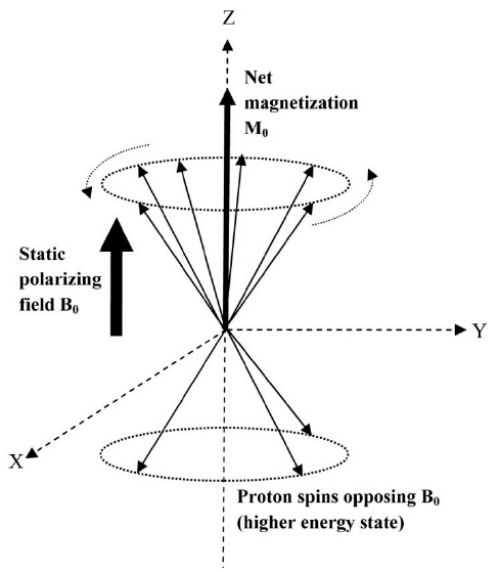


Figure 2: Net magnetization of a sample in an external magnetic field [3]

The equilibrium magnetization M_0 thus becomes

$$M_0 = \frac{N\gamma^2\hbar^2 I(I+1)}{3k_B T} \tag{4}$$

where N is the number of spins per unit volume.

1.2 Chemical shift

Nuclei in molecules are not fully isolated: their local environment has an effect on their resonance frequency. In addition to the applied external magnetic field, the nuclei also experience magnetic field contributions from the electron shells around them, resulting in a decrease in the experienced magnetic field strength (shielding) or in some cases, an increase.

The resulting field B_0' is described using the *shielding constant* σ as

$$B_0' = (1-\sigma)B_0$$

where B_0 is the original external field.

This shielding results in a shift of the resonance frequency. This effect is called the *chemical shift*, and is usually given terms of an easily measurable reference substance, as comparing it to resonant frequency of an isolated nucleus is not experimentally feasible. Thus, the chemical shift is defined as

$$\delta = \sigma_{ref} - \sigma = \frac{\omega - \omega_{ref}}{\omega_{ref}} \tag{5}$$

where ω_{ref} is the resonant frequency of the reference sample. The chemical shift is usually measured in the dimensionless unit *ppm* (parts per million).

The value of the chemical shift is fully determined by the environment, so a given set of chemical shift will be a characteristic of the molecular group that the measured nucleus is part of.

1.3 RF pulses and relaxation times

A non-equilibrium magnetization can be achieved by subjecting the sample to RF radiation at the resonant frequency, which flips some of the spins, thus changing the net magnetization. This is accomplished by applying short radio-frequency (RF) pulses perpendicular to the static magnetic field B_0, typically at the Larmor frequency $\omega_L = \gamma B_0$. These pulses rotate the macroscopic magnetization vector away from its equilibrium position along the z-axis.

After the RF pulse is turned off, the system gradually returns to thermal equilibrium through two relaxation processes. The *longitudinal relaxation time* T_1 characterizes the recovery of the z-component of the magnetization (M_z) due to energy exchange with the surrounding lattice (spin-lattice relaxation). The *transverse relaxation time* T_2 describes the loss of phase coherence among the spins in the transverse plane (M_{xy}), due to spin-spin interactions and local magnetic field inhomogeneities.

These relaxation times are essential for understanding NMR signal behavior and determine how quickly experiments can be repeated. Accurate measurement of T_1 and T_2 is crucial for quantitative analysis and for designing pulse sequences that maximize signal-to-noise ratio and spectral resolution.

1.4 Influence of a High-Frequency Field: 90° and 180° Pulses

In pulsed NMR, radio-frequency (RF) pulses are used to manipulate the orientation of nuclear spins relative to the static magnetic field B_0. A 90° pulse (or $\pi/2$ pulse) rotates the net magnetization vector from its equilibrium position along the z-axis into the transverse x-y plane, initiating precession at the Larmor frequency $\omega_L = \gamma B_0$. This transverse magnetization induces a signal in the receiver coil known as the free induction decay (FID).

The duration τ and amplitude B_{rf} of the RF pulse must satisfy $\gamma B_{\mathrm{rf}}\tau = \pi/2$ for a 90° rotation. Deviations from this condition result in incomplete rotation, reducing the transverse component and thus the signal intensity.

A 180° pulse (or π pulse) inverts the magnetization vector, flipping spins from $+1/2$ to $-1/2$ states and vice versa. The condition for a 180° rotation is $\gamma B_{\mathrm{rf}}\tau = \pi$. Such pulses are essential for techniques like Hahn Echo and Carr-Purcell-Meiboom-Gill (CPMG), where they are used to refocus dephased spins and counteract field inhomogeneities.

Pulse effectiveness depends critically on accurate calibration of pulse durations, RF field homogeneity, and matching the gyromagnetic ratio γ of the nuclei being observed.

1.5 Hahn Echo

The Hahn Echo sequence is a fundamental technique in NMR used to refocus dephasing caused by static magnetic field inhomogeneities. After a 90° pulse tips the magnetization into the transverse plane, spins at different spatial positions begin to precess at slightly different Larmor frequencies due to local variations in B_0, leading to dephasing and signal decay.

To counteract this, a 180° pulse is applied after a delay time τ. This pulse inverts the spin vectors about the axis of the RF field (typically the x-axis in the rotating frame), effectively reversing their relative phase evolution. After an additional delay τ, the spins re-align, forming a coherent signal called the Hahn echo at time $t = 2\tau$.

Mathematically, the echo amplitude $M(2\tau)$ is given by:

$$M(2\tau) = M(0) \exp\left(-\frac{2\tau}{T_2}\right) \exp\left(-\frac{1}{3}\gamma^2 G^2 D \tau^3\right),$$

where T_2 is the transverse relaxation time, G is the magnetic field gradient, D is the diffusion coefficient, and γ is the gyromagnetic ratio.

The Hahn Echo sequence isolates the intrinsic transverse relaxation (T_2) from additional broadening effects due to field inhomogeneity. However, in liquids, diffusion through magnetic field gradients can attenuate the echo, especially at longer τ. This effect is exploited in diffusion measurements.

For improved accuracy, especially when many echoes are required (as in the CPMG sequence), precise RF pulse calibration and homogeneous B_{rf} fields are essential to avoid cumulative phase errors.

2 Experimental Setup

2.1 NMR Spectrometer Setup

In nuclear magnetic resonance (NMR), a spectrometer is used to investigate the magnetic properties of atomic nuclei. A typical NMR spectrometer consists of several main components: a coil, a strong magnet, an RF (radio frequency) generator, a receiver, and a computer for signal processing.

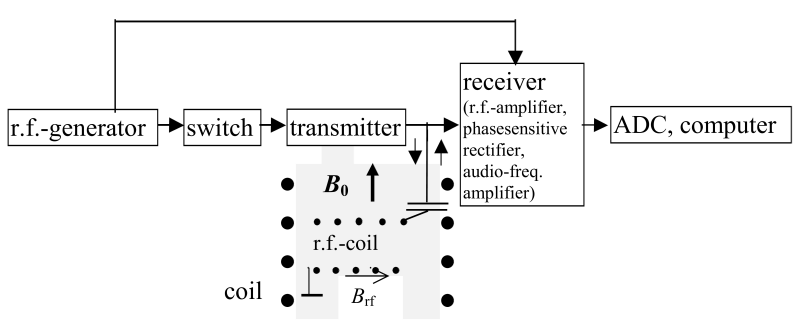


Figure 3: Simplified Experimental Set-up

The coil is part of a resonant circuit tuned to a specific frequency, which depends on the strength of the static magnetic field B_0 produced by a superconducting magnet. An RF generator produces electromagnetic pulses at or near the Larmor frequency, which is close

to the resonance frequency of the coil. When these RF pulses are applied, they generate a transverse magnetic field B_{rf} inside the coil.

This B_{rf} field interacts with the magnetic moments of the nuclei in the sample, exciting them from equilibrium and causing them to precess. As a result, the excited nuclei emit a response signal, which is detected by the same coil now acting as a receiver.

This induced signal, referred to as the free induction decay (FID), is converted into digital form by an analog-to-digital converter and sent to a computer. The computer then processes the time-domain signal using Fourier transformation to generate a frequency-domain NMR spectrum.

Pulsed NMR spectrometers use short, controlled RF pulses instead of continuous waves. This allows fine control over parameters such as pulse width, amplitude, timing, shape, and phase, enabling a wide variety of experimental techniques. For studying liquids, an RF field strength of about 12 mT is sufficient, while solid-state samples typically require stronger RF fields and shorter pulse durations.

The spectrometer operates at a fixed frequency and static field, and signal detection begins after the RF pulse ends. However, electronic transients and ring-down effects introduce a short dead time immediately after the pulse, during which the receiver cannot accurately record signals. While this limits the observation of very short relaxation signals, it does not affect spin echo experiments, which allow delayed signal refocusing.

Phase-sensitive detection ensures that the received signal retains information about the relative phase between the nuclear magnetization and the transmitted RF signal. This phase modulation is preserved in the time domain and is essential for accurate spectral interpretation after transformation into the frequency domain.

3 Analysis and Results

3.1 Task 1: Determination of the 90° ($\pi/2$) and 180° (π) H-pulse lengths using a water sample

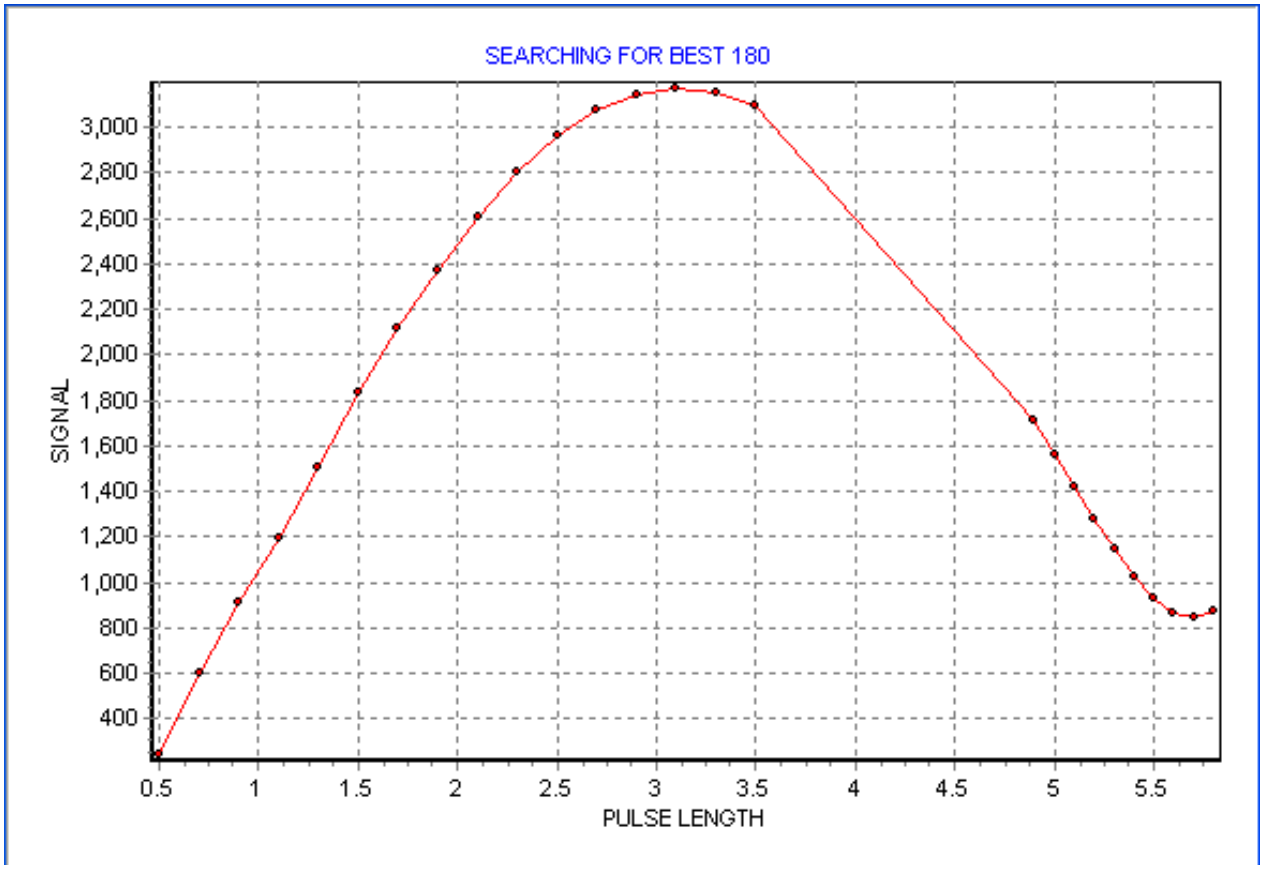


Figure 4: Signal measured against pulse length, plotted using the RINMR software.

The pulse length was calibrated automatically within the RINMR software using its built-in function, which was used frequently between measurements. The 90° pulse (P90) was typically set to approximately 2.9 μs, and the 180° pulse (P180) to 5.9 μs, consistent with the peak positions visible in Figure 4.

The signal intensity shows a sinusoidal dependence on pulse length, following the relation $\sin(\gamma B_1 t_p)$, where γ is the gyromagnetic ratio, B_1 is the RF field strength, and t_p is the pulse duration. The maximum signal is observed at $t_p = 2.9$ s, corresponding to a 90° pulse, as this rotation places the net magnetization fully in the transverse plane. The first minimum appears at $t_p = 5.9$ s, corresponding to a 180° pulse, where the magnetization is inverted and no transverse component remains to generate a detectable signal.

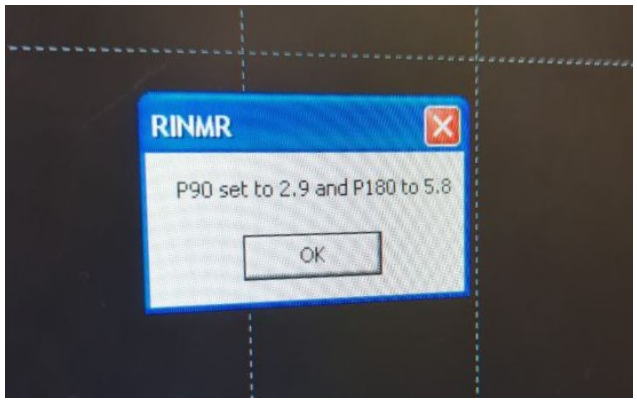


Figure 5: Calibrated Pulse lengths

3.2 Task 2: Determination of T1-relaxation times in other samples

The longitudinal relaxation time T_1 characterizes how quickly nuclear spins realign with the external magnetic field after being disturbed by an RF pulse.

To measure T_1, we first apply a π pulse to flip the magnetization to the $-z$ direction. Then we wait for a time τ, during which the magnetization starts to recover along the z-axis. After that, we apply a $\pi/2$ pulse to bring the current magnetization into the transverse plane so we can detect it (machine cannot detect magnetization when the spins are aligned with the z axis). By repeating this process with different τ values, we see how fast the magnetization recovers, and from that, we determine the T_1 relaxation time.

$$M_z(\tau) = M_0 \left(1 - e^{-\tau/T_1}\right),$$

where M_0 represents the equilibrium magnetization. Selecting a maximum τ of about five times T_1 ensures reliable curve fitting without unnecessary measurement time.

In the current experiment, the $\pi - \tau - \pi/2$ pulse sequence was implemented. After each acquisition, a relaxation delay was introduced to allow full recovery before repeating the measurement. For each delay value τ between 0 and 3 seconds, sixteen measurements were averaged to improve data consistency for accurate determination of T_1.

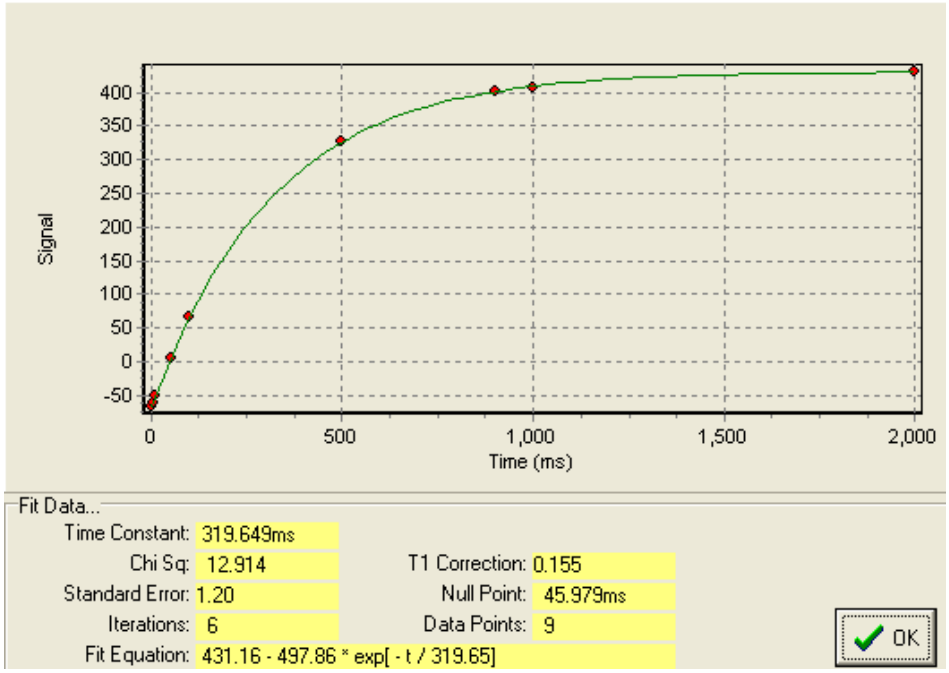


Figure 6: Water

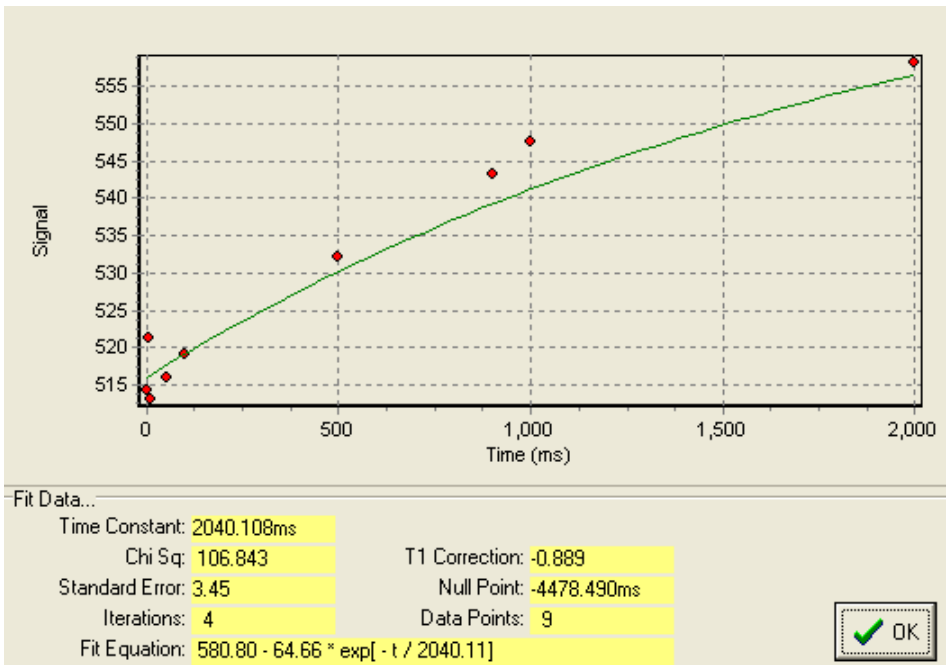


Figure 7: Sample 2

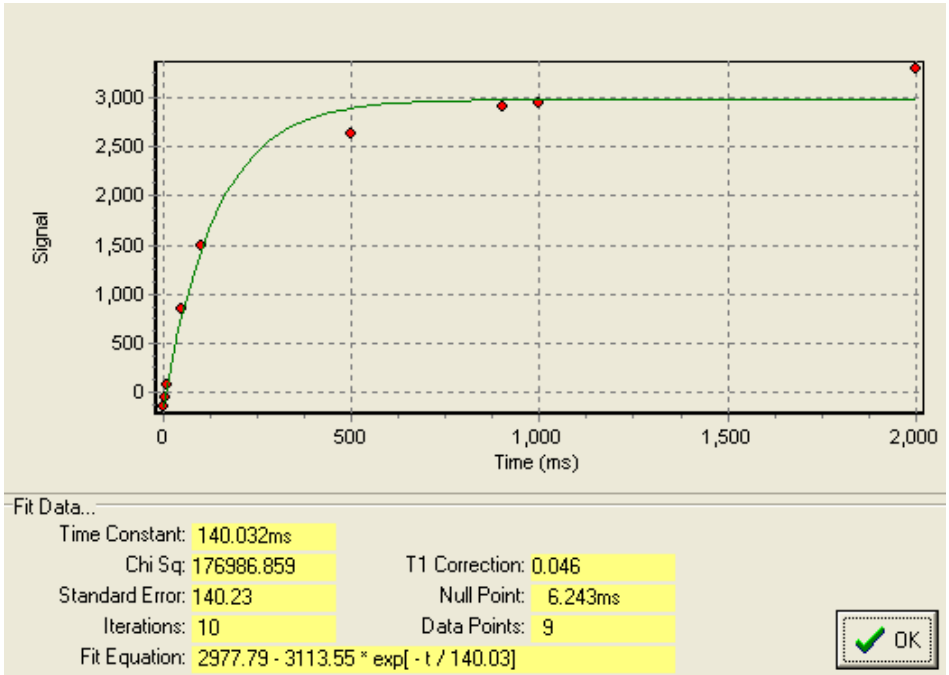


Figure 8: Sample 5

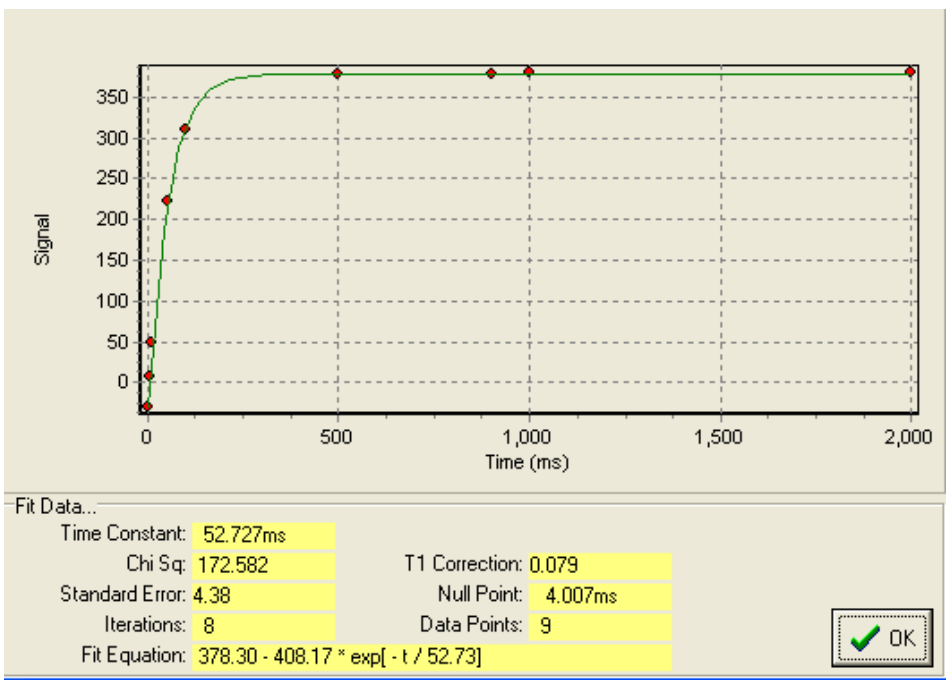


Figure 9: Oil

The measured T_1 values are generally within the expected range for the respective substances. The low T_1 value of the oil sample at 52.7 ms is consistent with its higher molecular weight and increased viscosity, which lead to faster spin-lattice relaxation. In contrast, the T_1 value

Sample	$\boldsymbol{T_1}$ **(ms)**
Water (Sample 1)	319.549
Tap Water (Sample 2)	2040.108
n-Butyl Alcohol (Sample 5)	140.032
Oil (assumed Hexamethyldisiloxane)	52.727

Table 1: Measured longitudinal relaxation times T_1 for various NMR samples using the inversion recovery method.

for tap water appears unusually high at 2040 ms; however, this can be attributed to a poor fit of the recovery curve, making the result less reliable. The values obtained for distilled water and n-butyl alcohol, 319.5 ms and 140.0 ms respectively, are reasonable.

Fast-tumbling molecules, such as water, interact less efficiently with their surroundings, resulting in weaker coupling and a longer T_1 relaxation time. In contrast, slow-moving or more viscous molecules, like oils and gels, exchange energy more effectively with their environment, leading to stronger coupling and therefore a shorter T_1.

3.3 Task 3: Measurement of the Free Induction (FID) of a Water Sample and Determination of the Time Constant $T_{2\text{FID}}$ of the Exponential Drop of the NMR Signals as well as the Line Width $\Delta\nu$ of the NMR Signals after the Fourier Transformation

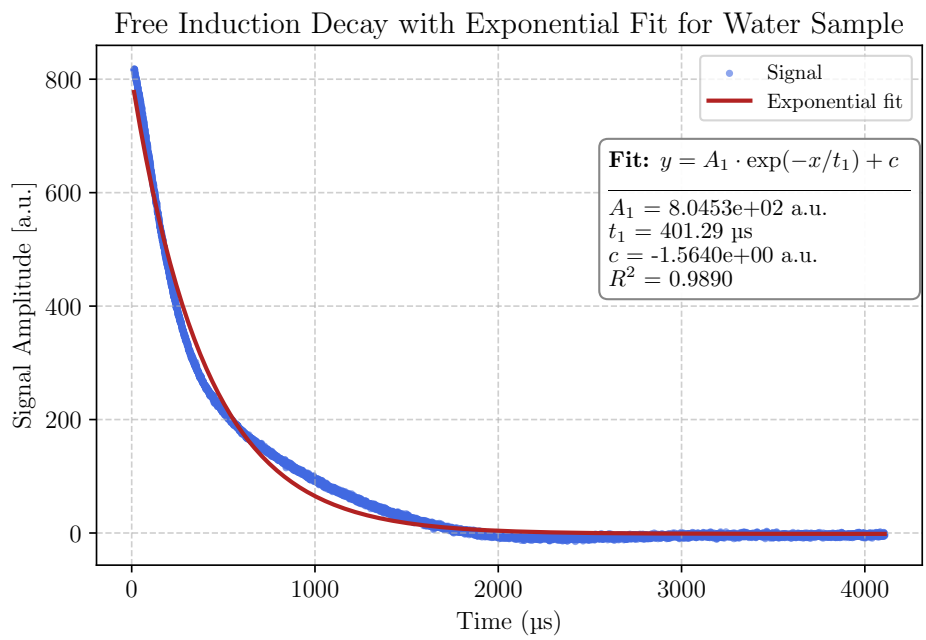


Figure 10: Water Sample FID

The raw data is exported without units in three columns. The first column is time in seconds. The second column is the amplitude [a.u.] of the real part of the signal which is used to plot the FID graph.

The parameter t_1 in the FID fit corresponds to $T_{2,\text{FID}}$. We obtain the value $T_{2,\text{FID}} = 401.29$ ms, which lies within the expected range for a water sample. The fit yields a reasonably high coefficient of determination, $R^2 = 0.989$, indicating a good fit.

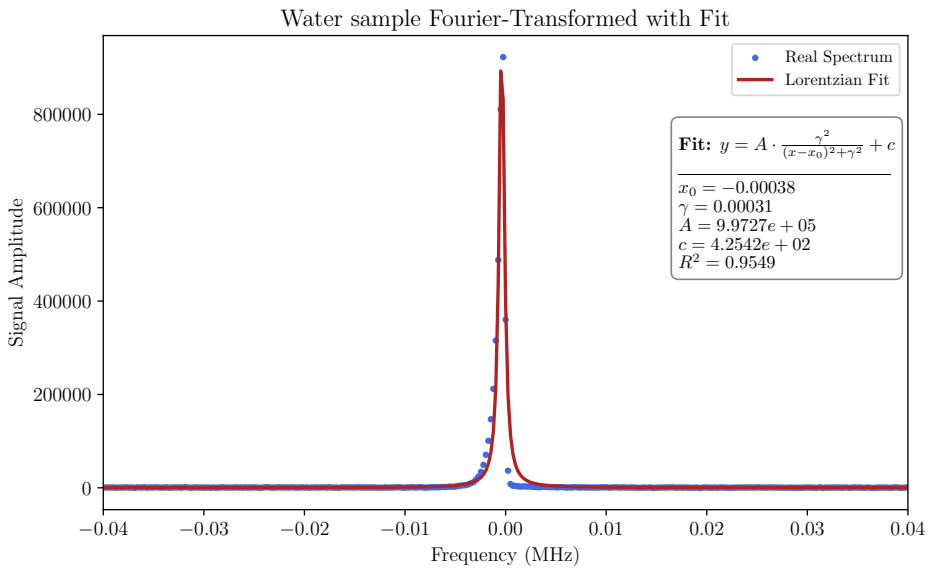


Figure 11: Water Sample FID Fourier-Transformed

The data was exported from the software without any units on the axes. For the Fourier-transformed graph, we deduce the x-axis units based on Task 4. In that task, we manually applied a frequency shift of 10 kHz, which corresponds to a value of 0.01 along the exported data's x-axis. From this, we infer that the unit of the x-axis is megahertz (MHz), since 0.01 MHz = 10 kHz.

For the Fourier-transformed graph, we deduce the x-axis units from Task 4. The shift we applied manually in Task 4 is 10 kHz, and from that we can deduce the data was exported in MHz. We did a Lorentzian fit over the Fourier-transformed data.

$$\Delta\nu_{12} = 2\gamma$$

$$T_2 = \frac{1}{2\pi\gamma}$$

With $\gamma = 0.00031$ MHz = 310 Hz, we obtain:

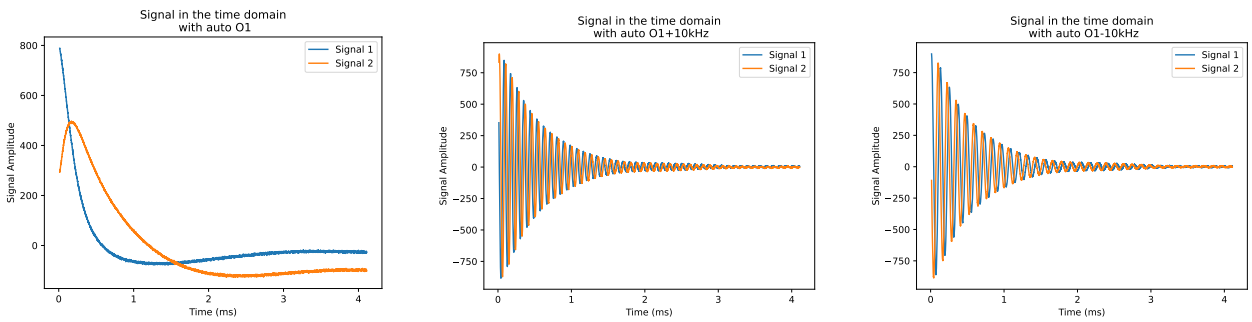
$$T_2 = \frac{1}{2\pi \cdot 310 \text{ Hz}} \approx 0.000513 \text{ s} = \boxed{513 \text{ µs}}$$

We extracted the transverse relaxation time T_2 from both the exponential decay of the FID and the Fourier-transformed spectrum. The time-domain fit yielded $T_{2,\text{FID}} \approx 401$ µs, while the Lorentzian linewidth gave $T_2 \approx 513$ µs. The discrepancy arises due to magnetic field inhomogeneities, which accelerate dephasing in the FID. As expected, the linewidth-based method yields a slightly longer T_2, but both values remain lower than the true relaxation time, which can only be accurately determined using refocusing techniques such as Hahn echo.

3.4 Task 4: Estimating the uncertainty in chemical shift measurements

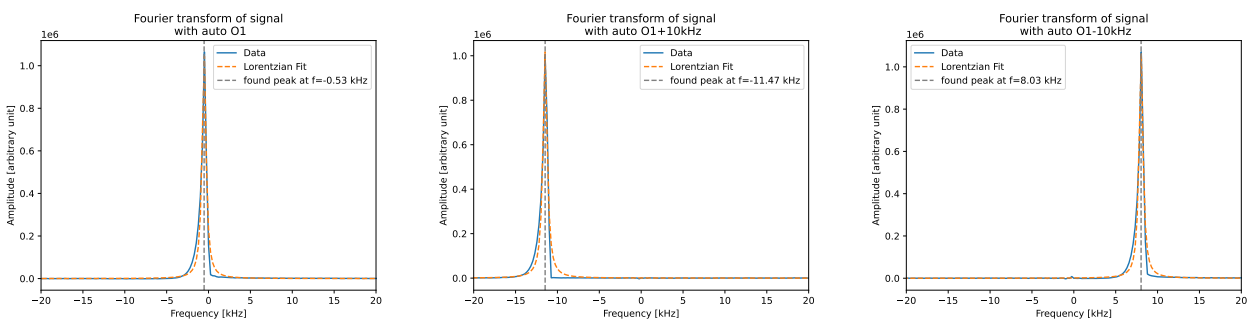
As mentioned in Section 1.2, chemical shift measures the shift in resonance frequency compared to a reference sample. In this task, we estimate the accuracy of these measurements.

For this, we first take a measurement with the offset frequency set automatically by the program at O1=344698 Hz, then subsequently take measurements for this value increased and decreased manually by 10 kHz. The transmitter frequency is equal to the sum of the spectrometer frequency (SF) and the offset frequency (O1). Ideally, we expect that changing the offset frequency would result in a shift of the resonant frequency found in the Fourier domain by exactly the shift in the frequency.



(a) O1=344698 Hz (b) O1=354698 Hz (c) O1=334698 Hz

Figure 12: The recorded NMR signals in the time domain with different offset frequencies



(a) O1=344698 Hz (b) O1=354698 Hz (c) O1=334698 Hz

Figure 13: The recorded NMR signals in the frequency domain with different offset frequencies

In the time domain the two signals correspond to the two perpendicular components of the induced current, or equivalently, the real and the imaginary parts of the signal. These come from coils that detect the transverse magnetization (in the plane perpendicular to the static B-field), but from two orthogonal directions. This allows for a true frequency spectrum after the Fourier transform, with the correct phases and signs.

At the auto offset frequency the transmitter frequency should correspond exactly to the Larmor frequency of the nuclei we are observing. In the rotating frame the spins appear stationary, so there in no frequency difference: the signal is purely a decaying exponential (no oscillation) and the fourier peak is at 0. This is exactly what we see in Fig.12a and Fig. 13a.

By changing the offset frequency, the spins no longer appear stationary in the rotating frame, but precess with the frequency of the change in the offset frequency. This results in an oscillatory signal with exponentially decaying amplitude in the time domain, and a peak shifted from zero by the change in offset frequency in the frequency domain. This is what we see for an increased offset by 10 kHz (Fig. 12b and Fig. 13b) and for a decreased offset by the same amount (Fig. 12c and Fig. 13c). In the following table the obtained peaks are compared to their expected values:

Obtained peak [kHz]	**Expected [kHz]**	**Absolute error [kHz]**	**Error [ppm]**
-0.53	0.00	0.53	22.53
-11.47	10.00	1.47	62.75
8.03	-10.00	1.97	84.09

Table 2: Comparison of obtained peak frequencies with expected values.

The ppm error in the above table is calculated using the spectrometer frequency of 23.4 MHz and using equation 5. Typical values of chemical shift are between 0-10 ppm[4], so this resolution is not enough for proper chemical shift measurements.

Remark: We used the fact that the shift in the frequency for a different offset frequency has to match the offset itself to determine the units of the recorded data: frequencies were in MHz, while the time was given in μs.

3.5 Task 5: Hahn echo of a Water sample and Sample 5

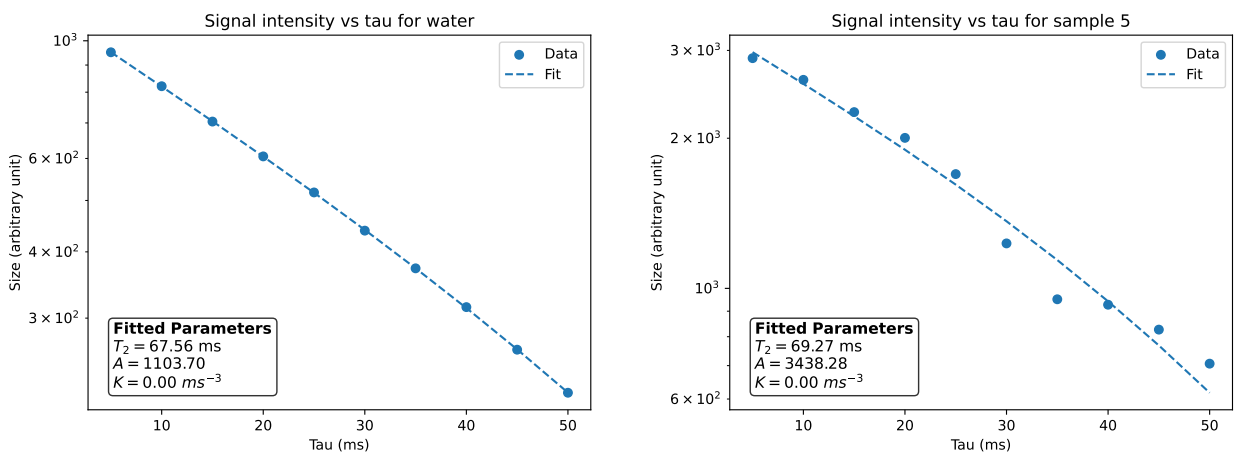
The Hahn echo sequence involves applying two radiofrequency pulses with specific phase relationships to refocus dephasing spins, resulting in the formation of an "echo" signal. The Hahn echo is especially significant for spin-$\frac{1}{2}$ nuclei [5]. The Hahn echo corrects for dephasing of nuclear spins caused by inhomogeneities in the magnetic field. These inhomogeneities cause individual spins to precess at slightly different frequencies, leading to signal decay even when the intrinsic spin–spin relaxation T_2 hasn't fully acted.

In this technique, a $\pi/2$ pulse is applied, followed by a time delay τ, and then a π pulse is applied. This sequence leads to the rephasing of spins and the appearance of an echo in the signal with a maximum at $t = 2\tau$. The amplitude of the echo decreases with the delay time τ: the longer you wait between pulses, the more transverse magnetization is lost due to irreversible spin–spin interactions (T_2 relaxation), not due to field inhomogeneity which the echo corrects for. This decay can be described by an exponential relation of the form

$$S = Aexp\left(-\frac{2\tau}{T_2} - \frac{K}{\tau^3}\right) \tag{6}$$

where S is the signal intensity, τ is the delay time between the two pulses, T_2 is the spin-spin relaxation time, A is arbitrary fitting constant, and K-term accounts for other kind of decays that are not because of magnetic field inhomogeneity or T_2 relaxation (eg. diffusion).

For this task, we measured the Hahn echo of a water sample and an n-butanol sample (Sample 5) for ten different values of the delay time τ. We could then use the recorded intensities and fit equation 6 to find the T_2 relaxation time.



(a) water

(b) n-butanol

Figure 14: Measured amplitude of the Hahn echo as function of τ for water and for n-butanol fitted with the function form eq. 6

From the fits, we determined the T_2 relaxation times: 67.56 ms for water, and 69.27 ms for n-butanol. In both cases the constant K tuned out to be 0 at least to 4 decimals, which indicates that the decay in amplitude is almost completely determined by T_2 relaxation.

Comparing to Lorentzian fit of Task 3, the T_2 relaxation time for water is larger when using the Hahn echo. This can be explained by the fact that the Hahn sequence takes care of the effects of inhomogeneities in the magnetic field, while the Free Induction Decay signal does not, which results in a faster decay. This means that the T_2 determined by the Hahn echo is closer to the true value.

3.6 Task 6: CPMG of a Water sample and Sample 5

When the refocusable decay (as discussed in Task 5) occurs more slowly than the rapid signal loss in the free induction decay, it becomes possible to capture multiple echo signals in succession before spin-lattice relaxation occurs. This principle underlies the Carr-Purcell Meiboom-Gill (CPMG) pulse sequence.[6] In its simplest form, the sequence of pulses consists of a Hahn echo pulse at $t = \tau$ followed by a series of refocusing π pulses in intervals of 2τ, resulting in successive echoes in the signal.

After applying the π-pulse n times and creating n echoes, the amplitude of each echo is obtained using equation 6 with τ replaced by $2n\tau$. This can then be used to determine T_2.

For this task, we used the CPMG pulse sequence for a water sample and Sample 5 (n-butanol). We adjusted the number N of pulses/echoes (parameter NECH in software) and the time interval τ (parameter TAU) until we got a sufficient decay in amplitude, and used the built-in feature of the software RINMR to determine T_2.

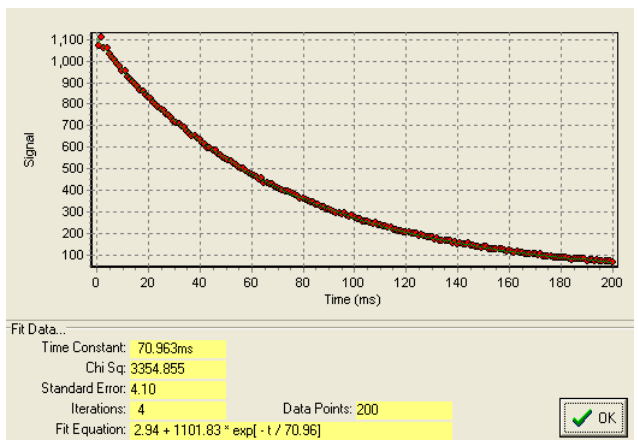
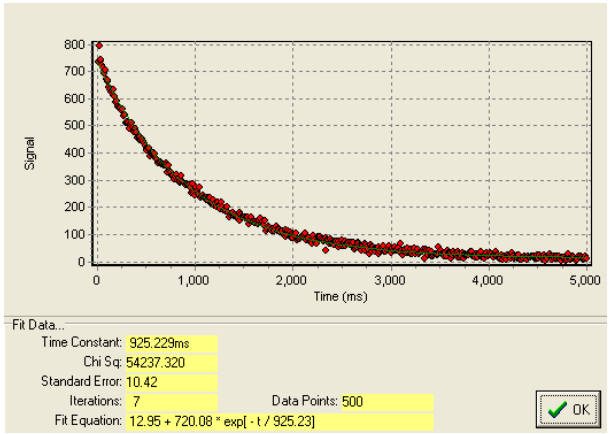
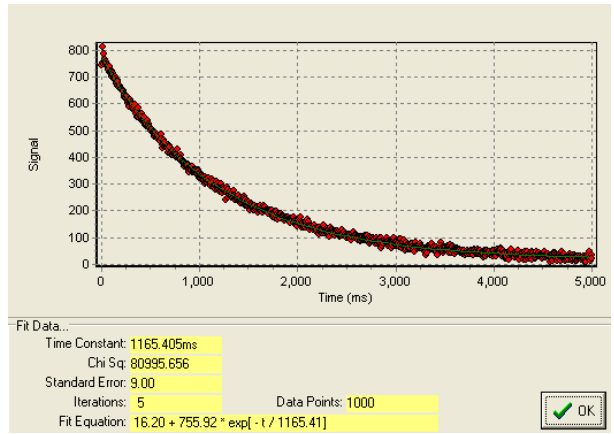


Figure 15: CPMG decay of water sample with τ=0.5 ms and N=200

For water, we obtained the value of $T_{2,CPMG} = 70.96$ ms which is close to the value of $T_{2,Hahn} = 67.56$ ms (see Fig. 15), but somewhat longer: a CPMG sequence refocuses reversible dephasing in a more efficient and repeated manner, getting rid of the influence of the inhomogeneities in the magnetic field better. So the value obtained with the CPMG sequence is much closer to the true value of T_2.



(a)



(b)

Figure 16: CPMG decay of Sample 5 with τ=5 ms and N=500 (left) and τ=2.5 ms and N=1000 (right)

For Sample 5 (n-butanol) we took measurements for two sets of parameters: once with τ=5 ms and N=500, then with τ=2.5 ms and N=1000. This resulted in covering the same total time interval of 5000 ms, but with different spacing of pulses. We obtained the values of $T_{2,CPMG}$ = 925.23 ms and 1165.41 ms, respectively. Both are much larger than the value of $T_{2,Hahn}$ = 69.27 obtained with a simple Hahn echo. This could mean that n-butanol experiences stronger dephasing due to field inhomogeneities and diffusion, which the CPMG sequence corrects for much more efficiently. We can also see that for smaller time interval between pulses the T_2 value increases. This could be because refocusing happens more frequently, so the spins have shorter time to dephase and speed the decay.

4 References

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